Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

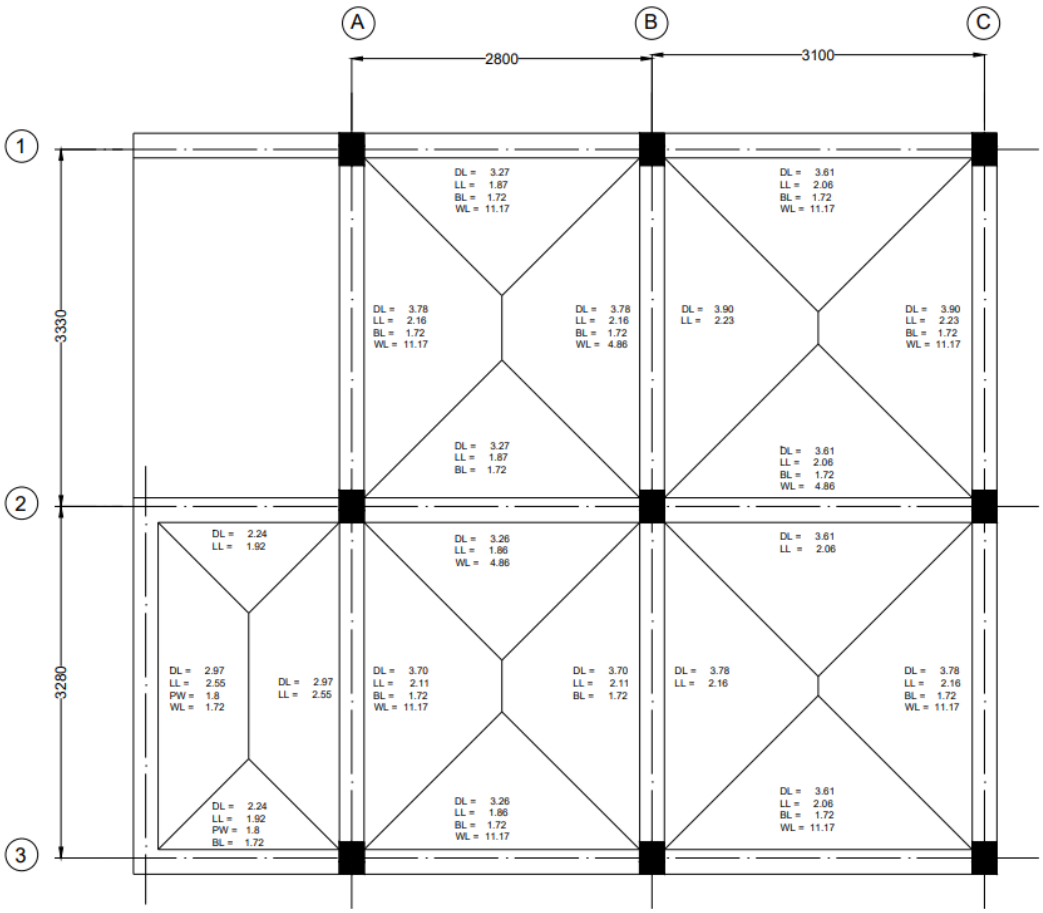


Figure 1: Beam Layout

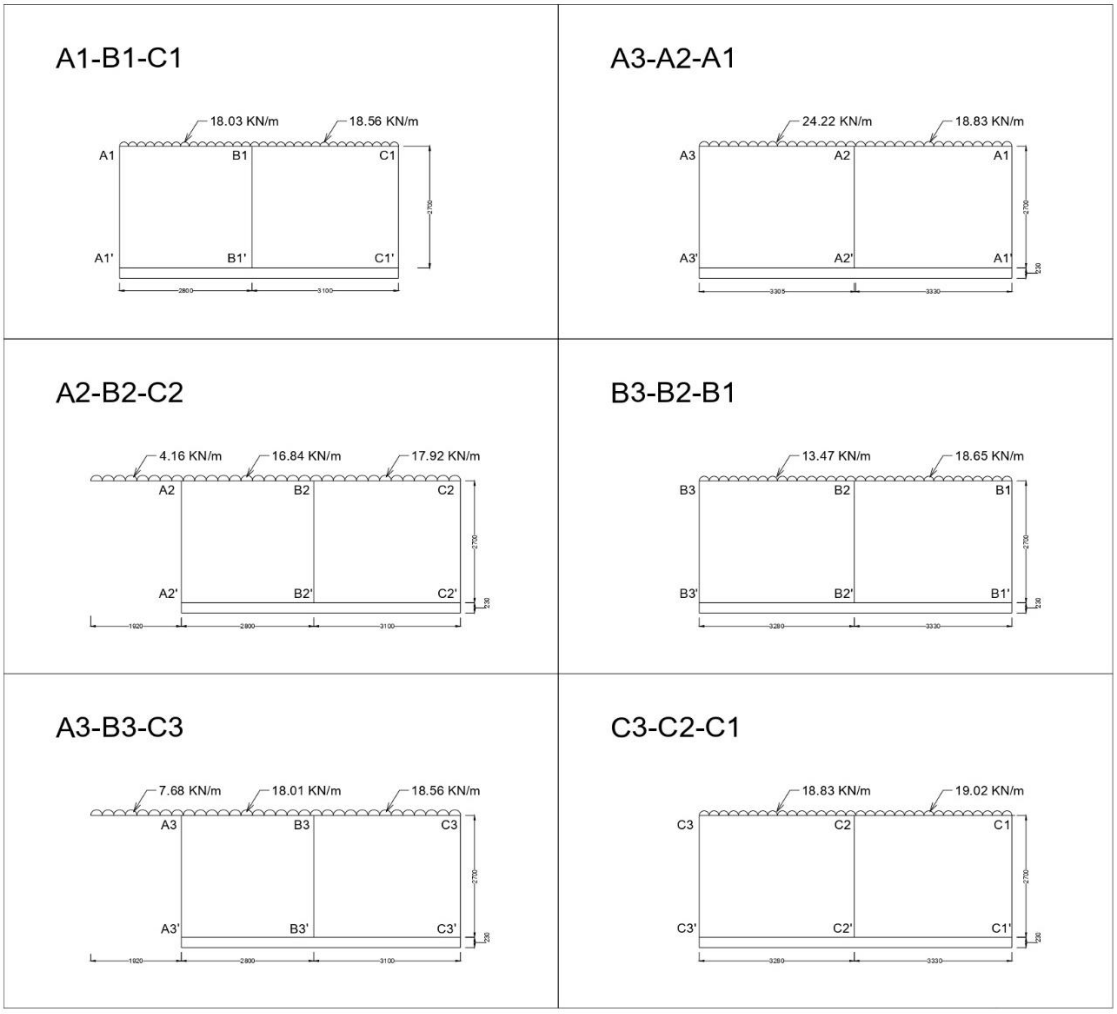
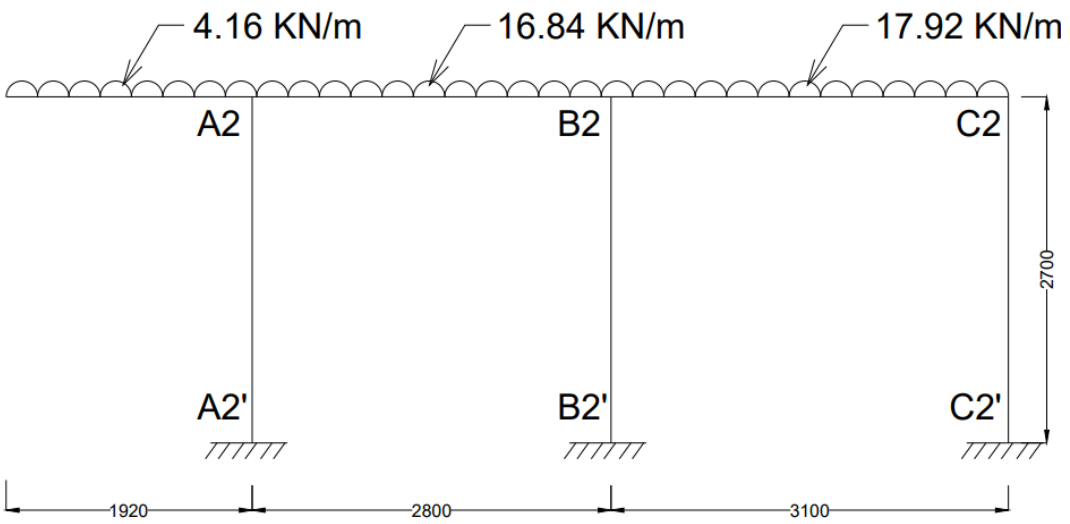


Figure 2: Frames.

Analysis of Frame A2-B2-C2



Step 1: Fixed End Moments

$MF_{Overhanging} = \frac{WL^2}{2} = \frac{4.16 \times 1.92^2}{2} = 7.67$ KN-m

$MF_{A2B2} = \frac{-WL^2}{12} = \frac{-16.84 \times 2.8^2}{12} = -11.00$ KN-m

$MF_{B2A2} = \frac{WL^2}{12} = \frac{-16.84 \times 2.8^2}{12} = +11.00$ KN-m

$MF_{B2C2} = \frac{-WL^2}{12} = \frac{-17.92 \times 3.1^2}{12} = -14.35$ KN-m

$MF_{C2B2} = \frac{WL^2}{12} = \frac{17.92 \times 3.1^2}{12} = +14.35$ KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{A2A2'} = MF_{A2'A2} = MF_{B2B2'} = MF_{B2'B2} = MF_{C2C2'} = MF_{C2'C2} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
A2	A2B2	1.428EI	2.909EI	-0.245
	A2A2′	1.481EI		-0.255
B2	B2A2	1.428EI	4.200EI	-0.170
	B2C2	1.290EI		-0.154
	B2B2′	1.481EI		-0.176
C2	C2B2	1.290EI	2.771EI	-0.233
	C2C2′	1.481EI		-0.267

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A2B2} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2A2} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{B2C2} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2B2} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{A2B2} = -\frac{1}{2}\left(\frac{1.428EI}{2.909EI}\right) = -0.245$

$R.F_{A2A2'} = -\frac{1}{2}\left(\frac{1.481EI}{2.909EI}\right) = -0.255$

$R.F_{B2A2} = -\frac{1}{2}\left(\frac{1.428EI}{4.200EI}\right) = -0.170$

$R.F_{B2C2} = -\frac{1}{2}\left(\frac{1.290EI}{4.200EI}\right) = -0.154$

$R.F_{B2B2'} = -\frac{1}{2}\left(\frac{1.481EI}{4.200EI}\right) = -0.176$

$R.F_{C2B2} = -\frac{1}{2}\left(\frac{1.290EI}{2.771EI}\right) = -0.233$

$R.F_{C2C2'} = -\frac{1}{2}\left(\frac{1.481EI}{2.771EI}\right) = -0.267$

Sum of Stiffness $\sum k$ at Joints

A2: 1.428EI + 1.481EI = 2.909EI

B2: 1.428EI + 1.290EI + 1.481EI = 4.200EI

C2: 1.290EI + 1.481EI = 2.771EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	A2A2′	1.481EI	4.444EI	-0.5
	B2B2′	1.481EI		-0.5
	C2C2′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{A2A2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{B2B2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{C2C2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

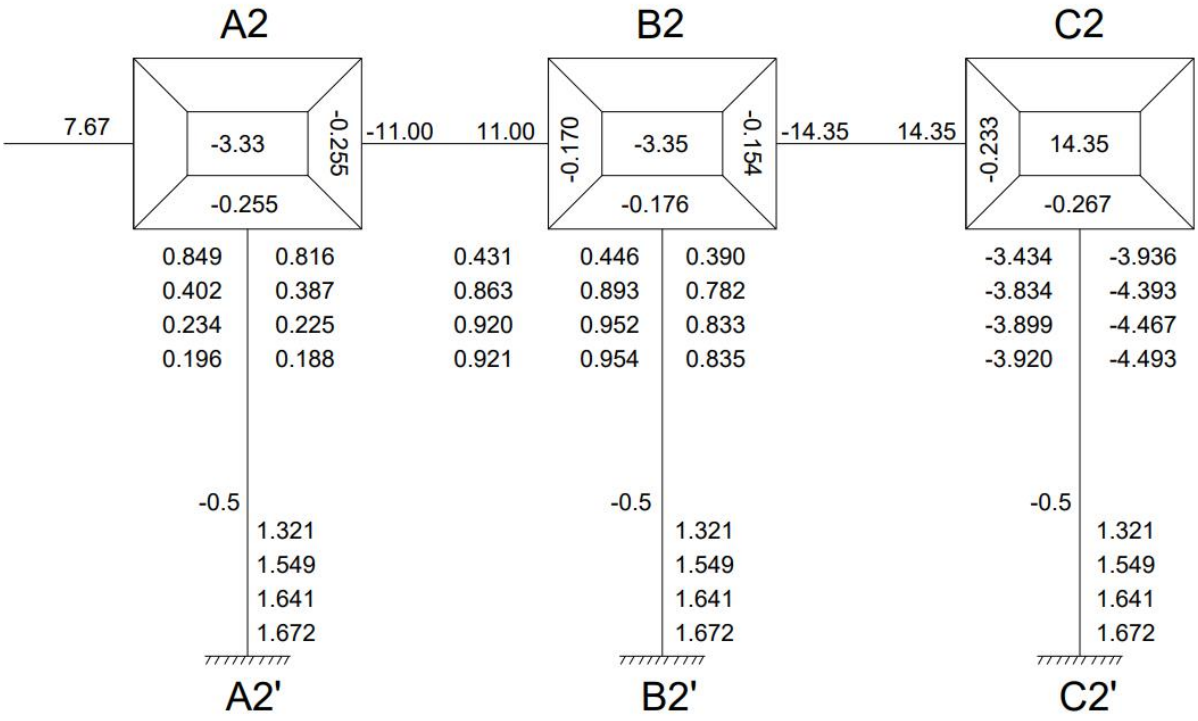
Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution

starting FEM at Joint A2 ∑FEM will be the sum of the overhang and the beam:

$\sum FEM_{A2} = 7.67 + (-11.00) = -3.33KN\text{-}m$



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

$M_{A2B2} = -0.245 \times [-3.33 + 0 + 0] = 0.816$

$M_{A2A2'} = -0.255 \times [-3.33 + 0 + 0] = 0.849$

$M_{B2A2} = -0.170 \times [-3.35 + 0.816 + 0] = 0.431$

$M_{B2C2} = -0.154 \times [-3.35 + 0.816 + 0] = 0.390$

$M_{B2B2'} = -0.176 \times [-3.35 + 0.816 + 0] = 0.446$

$M_{C2B2} = -0.233 \times [14.35 + 0.390 + 0] = -3.434$

$M_{C2C2'} = -0.267 \times [14.35 + 0.390 + 0] = -3.936$

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'} = -0.5 \times (0.849 + 0.446 + (-3.936)) = 1.321$

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A2B2} = -0.245 × [-3.33 + 1.321 + 0.431] = 0.387

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.321 + 0.431] = 0.402

M_{B2A2} = -0.170 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.863

M_{B2C2} = -0.154 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.782

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.893

M_{C2B2} = -0.233 × [14.35 + 1.321 + 0.782] = -3.834

$M_{C2C2'}$ = -0.267 × [14.35 + 1.321 + 0.782] = -4.393

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.402 + 0.893 + (-4.393)) = 1.549

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A2B2} = -0.245 × [-3.33 + 1.549 + 0.863] = 0.225

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.549 + 0.863] = 0.234

M_{B2A2} = -0.170 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.920

M_{B2C2} = -0.154 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.833

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.952

M_{C2B2} = -0.233 × [14.35 + 1.549 + 0.833] = -3.899

$M_{C2C2'}$ = -0.267 × [14.35 + 1.549 + 0.833] = -4.467

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.234 + 0.952 + (-4.467)) = 1.641

Cycle 4

Joint Rotation $(C.V. = R.F \times [\sum FEM + NCV + FCV])$

M_{A2B2} = -0.245 × [-3.33 + 1.641 + 0.920] = 0.188

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.641 + 0.920] = 0.196

M_{B2A2} = -0.170 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.921

M_{B2C2} = -0.154 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.835

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.954

M_{C2B2} = -0.233 × [14.35 + 1.641 + 0.835] = -3.920

$M_{C2C2'}$ = -0.267 × [14.35 + 1.641 + 0.835] = -4.493

Displacement Contribution

For All Columns $(R.V. = D.F \times [\sum Rotations\ Contribution\ at\ Columns])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.196 + 0.954 + (-4.493)) = 1.672

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0) "for Joint A2, the Overhang Moment is a fixed constant of +7.67 KN-m"

M_{A2B2} = -11.00 + 2(0.188) + 0.921 = -9.70

M_{B2A2} = 11.00 + 2(0.921) + 0.188 = 13.03

M_{B2C2} = -14.35+ 2(0.835) + (-3.920) = -16.60

M_{C2B2} = 14.35 + 2(-3.920) + 0.835 = 7.35

Column (With Sway M″= 1.672)

$M_{A2A2'}$ = 0 + 2(0.196) + 0 + 1.672 = 2.06

$M_{A2'A2}$ = 0 + 2(0) + 0.196 + 1.672 = 1.87

$M_{B2B2'}$ = 0 + 2(0.954) + 0 + 1.672 = 3.58

$M_{B2'B2}$ = 0 + 2(0) + 0.954 + 1.672 = 2.63

$M_{C2C2'}$ = 0 + 2(-4.493) + 0 + 1.672 = -7.31

$M_{C2'C2} = 0 + 2(0) + (-4.493) + 1.672 = -2.82$

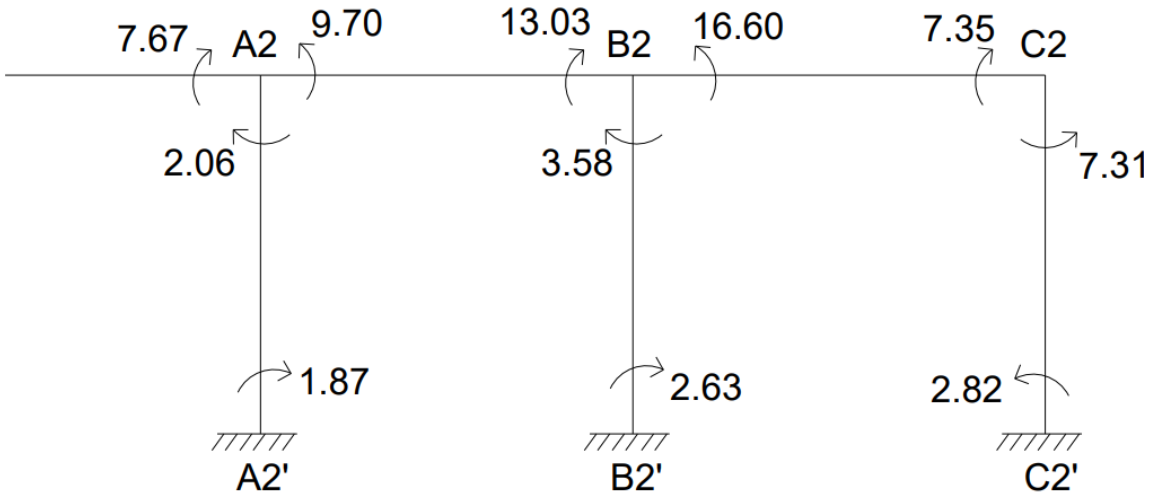
Equilibrium Check

"To check Joint A2, we must include that +7.67 overhanging moment along with the beam and column."

Joint A2 = $M_{overhanging} + M_{A2B2} + M_{A2A2'} = 7.67 + (-9.70) + 2.06 = 0.03 \approx 0$

Joint B2 = $M_{B2A2} + M_{B2C2'} + M_{B2B2'} = 13.03 + (-16.60) + 3.58 = 0.01 \approx 0$

Joint C2 = $M_{C2B2} + M_{C2C2'} = 7.35 - 7.31 = 0.04 \approx 0$



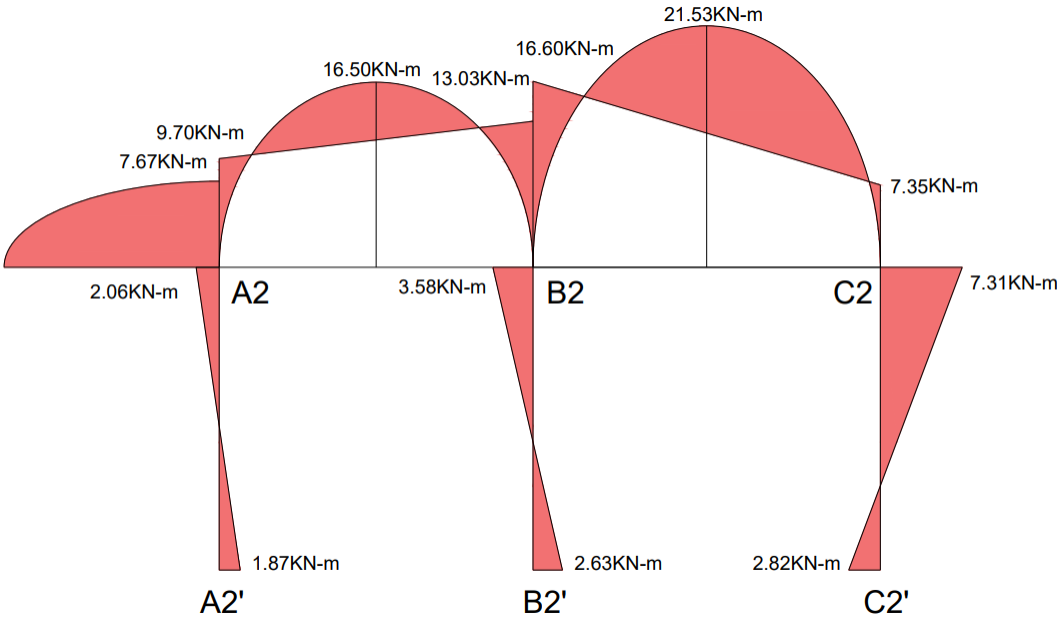
CALCULATION BENDING MOMENT

"Since the overhang is a cantilever, it doesn't have a simply supported "sagging" mid-span moment. Its moment is purely negative (hogging), starting at 0 at the free end and curving to 7.67 KN-m at support A2."

$BM_{A2B2} = \frac{WL^2}{8} = \frac{16.84 \times 2.8^2}{8} = 16.50KN\text{-}m$

$BM_{B2C2} = \frac{WL^2}{8} = \frac{17.92 \times 3.1^2}{8} = 21.53KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

From Overhang (Direct load transfer):

$R_{overhang} = w \times L = 4.16 \times 1.92 = 7.99KN$

For Span A2B2

Simply Supported Reactions:

R_{total} @ A2B2$= \frac{WL}{2} = \frac{16.84 \times 2.8}{2} = 23.58KN$

Reaction Due to End Moments: (Net clockwise moment means left support gets less load)

R_{moment} @ A2B2$= \frac{M_{A2B2} - M_{B2A2}}{L} = \frac{-9.70 + 13.03}{2.8} = -1.19KN$

Reaction Due to End Moments: (Net clockwise moment means right support gets more load)

R_{moment} @ A2B2$= \frac{M_{A2B2} - M_{B2A2}}{L} = \frac{+9.70 + 13.03}{2.8} = +1.19KN$

For Span B2C2

Simply Supported Reactions:

R_{total} @ B2C2= $\frac{WL}{2} = \frac{17.92 \times 3.1}{2}$ = 27.78KN

Reaction Due to End Moments: (Net clockwise moment means right support gets less load)

R_{moment} @ B2C2= $\frac{M_{B2C2} - M_{C2B2}}{L} = \frac{-16.60 + 7.35}{3.1}$ = -2.98KN

Reaction Due to End Moments: (Net clockwise moment means left support gets more load)

R_{moment} @ B2C2= $\frac{M_{B2C2} - M_{C2B2}}{L} = \frac{16.60 + 7.35}{3.1}$ = +2.98KN

Final Reactions:

$R_{A2} = R_{overhang} + R_{total}$ @ A2B2+ $R_{moments}$ @ $A2B2_{(Left\ Support)}$

R_{A2} =7.99 + 23.58 + (-1.19) = 30.38KN

$R_{B2} = (R_{total}$ @ A2B2+ $R_{moments}$ @ $A2B2_{(Right\ Support)}) + (R_{total}$ @ B2C2+ $R_{moments}$ @ $B2C2_{(Left\ Support)})$ =

R_{B2} = (23.58 + 1.19) + (27.78 + 2.98) = 55.53

$R_{C2} = R_{total}$ @ B2C2+ $R_{moments}$ @ $B2C2_{(Right\ Support)}$ =

R_{C2} = 27.78 + (-2.98) = 24.80 KN

